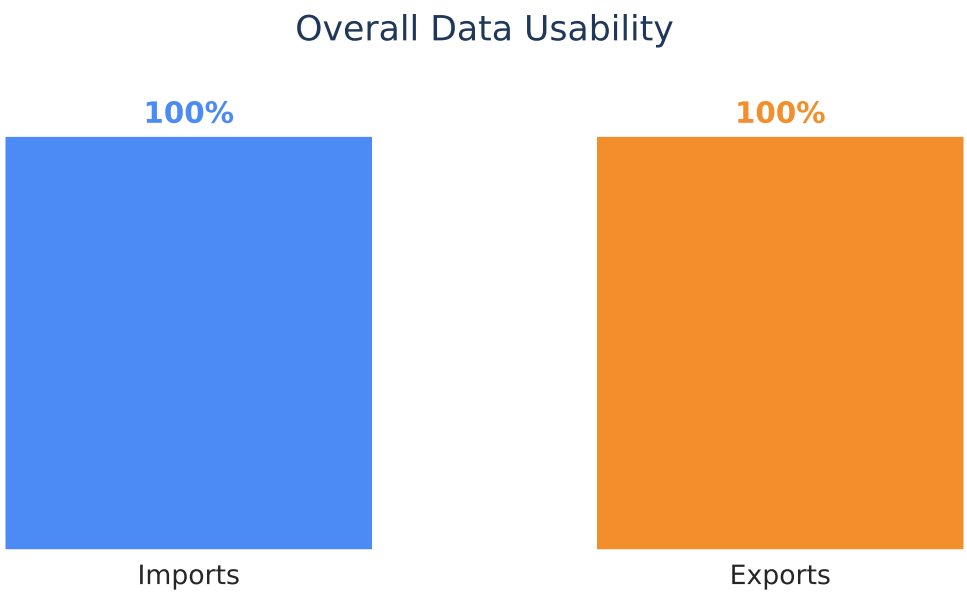
Trade Data Quality Diagnostic

Brazil ↔ Canada • 2024–2025

Understanding • Cleaning • Preparing for Analysis

Generated: 2026-02-17

Executive Summary



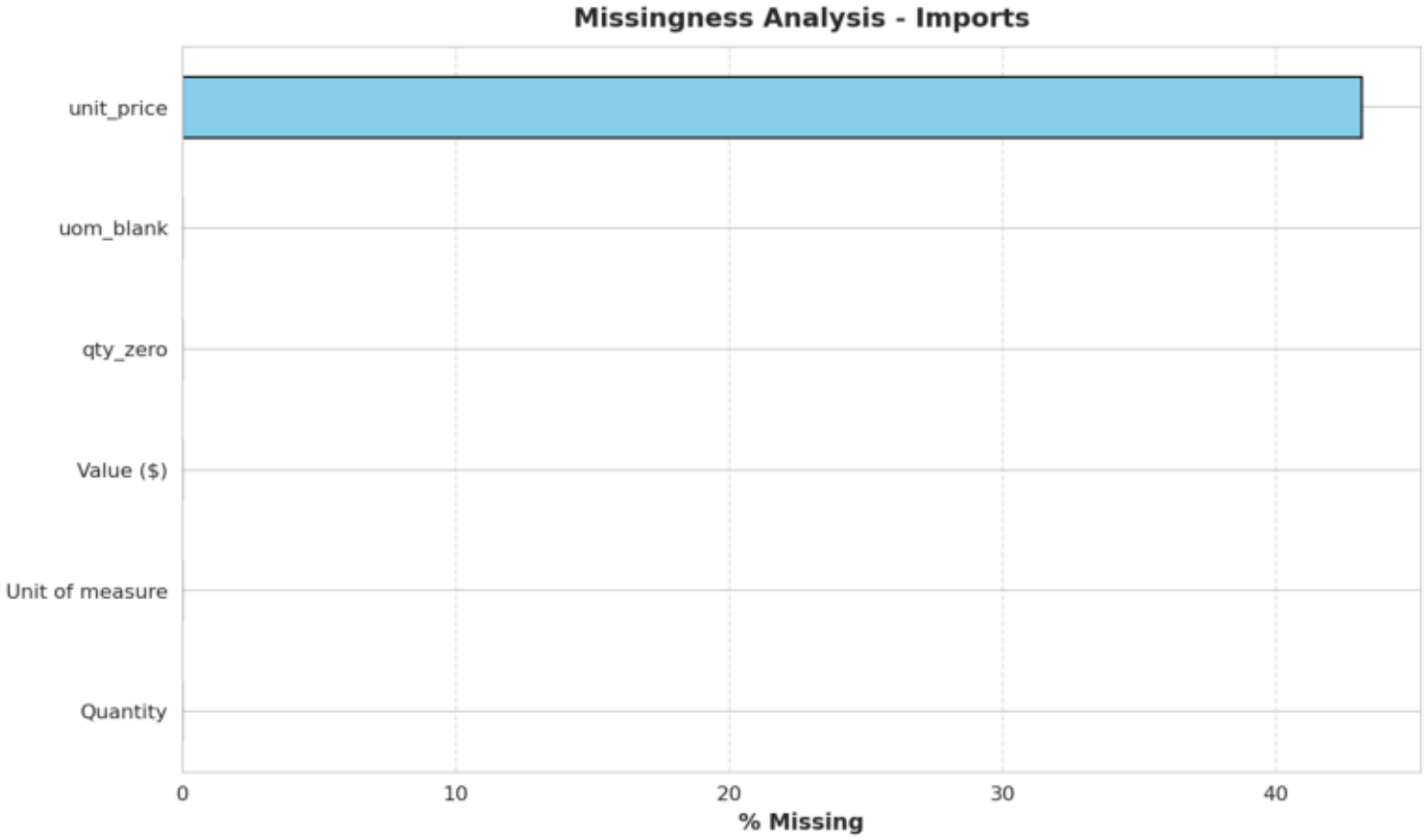
Where do we stand today?

• • Starting point: 63,716 import + 14,964 export records
• • After date cleaning: ~100.0% usable import rows, similar for exports
• • Zero-quantity records are the biggest obstacle — affecting price calculation in 43.1% to 63.5% of cases
• • Price outliers and inconsistent units appear in both directions
• • Temporal coverage is mostly complete — good foundation for trends

What should we do?

→ 1. Make a clear decision on zero-quantity records (impute / exclude / flag)
→ 2. Standardize units of measure per product group
→ 3. Flag or review price outliers before final reporting
→ 4. Normalize text (accents/special characters) for cleaner matching

Import Data — Key Patterns & Issues

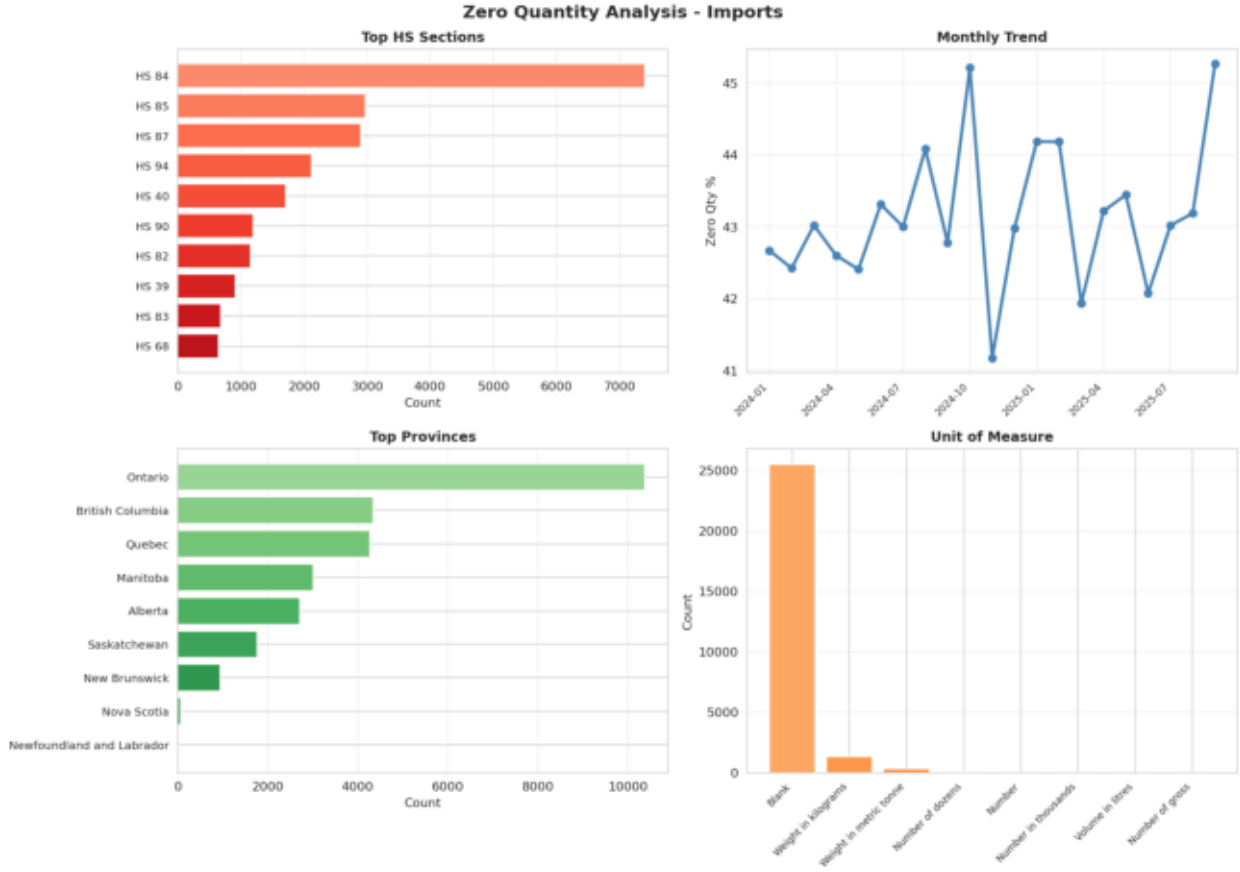


Missing & Invalid Data

- Quantity = 0 is the main reason unit_price cannot be calculated
- Core identifier fields (Period, Commodity, Country) are almost complete
- Very few truly missing values outside of derived fields

What should we do?

→ Treat zero-quantity rows as a special category rather than true missing data

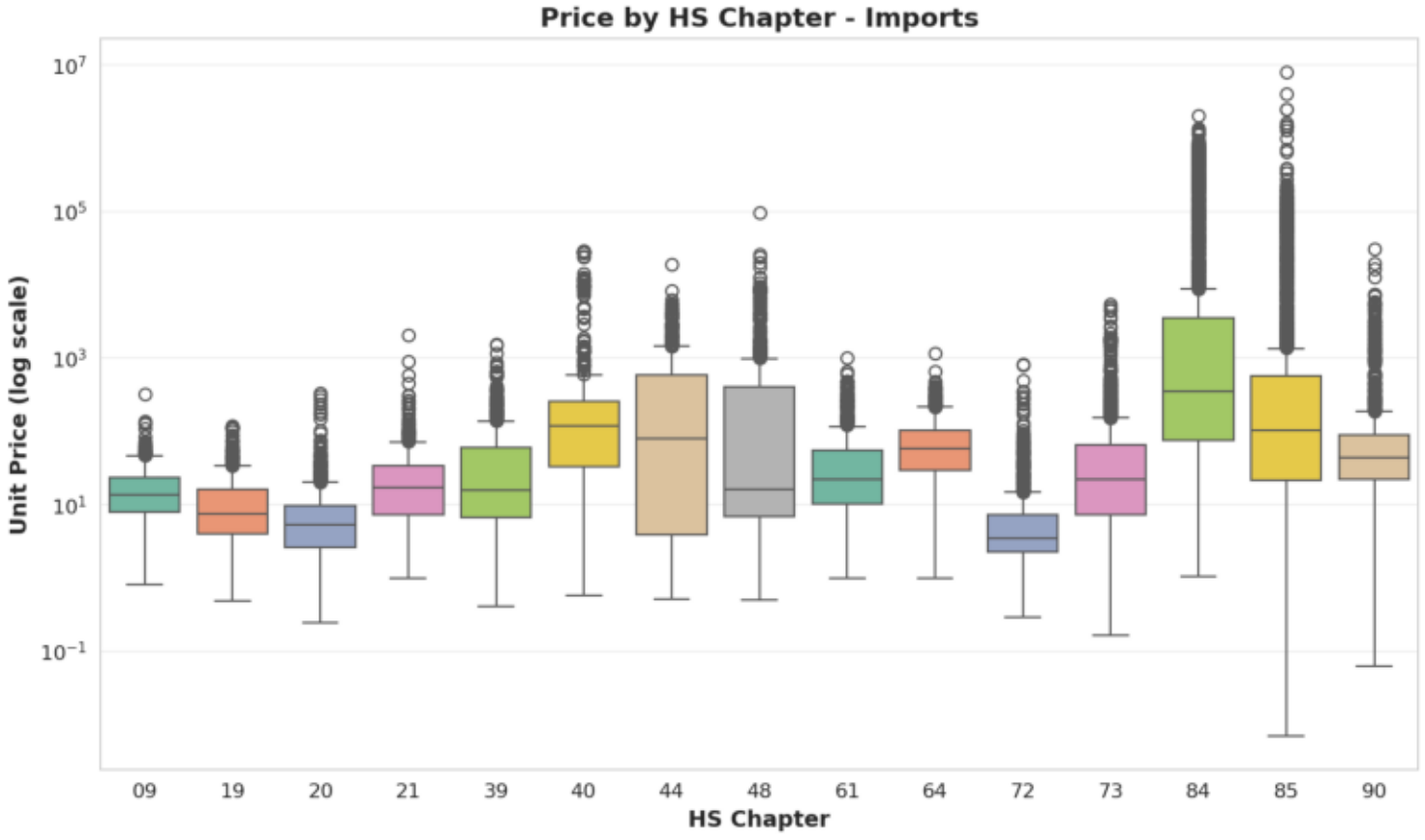


Zero Quantity Patterns

- 43.1% of import records show Quantity = 0
- This issue is consistent across months — not a seasonal artifact
- Certain HS sections (especially machinery & chemicals) show much higher rates

What should we do?

→ Most realistic options: (a) impute using median unit price per HS group, (b) exclude from unit price analysis



Price Behavior & Outliers

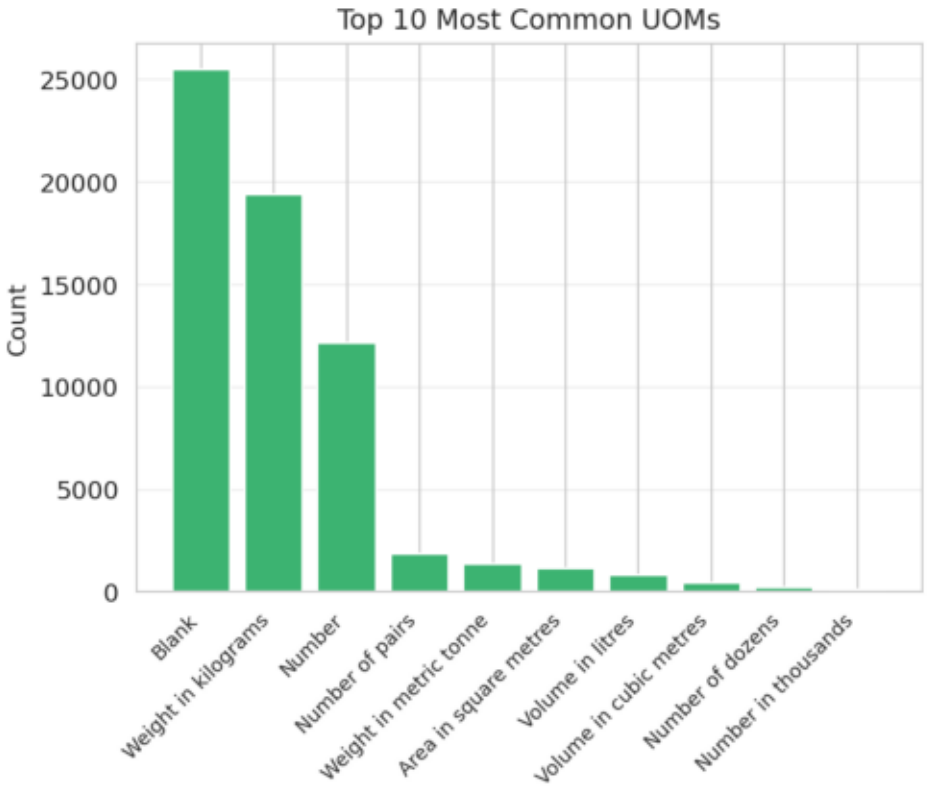
- Very wide price ranges within the same HS chapter — typical in trade data
- Electronics, machinery and vehicles show the highest median prices
- 4,235 category-specific outliers detected

What should we do?

→ Use category (HS2) specific outlier thresholds — global rules will flag too many legitimate high-value items

Unit of Measure Analysis - Imports

No Multi-UOM Products



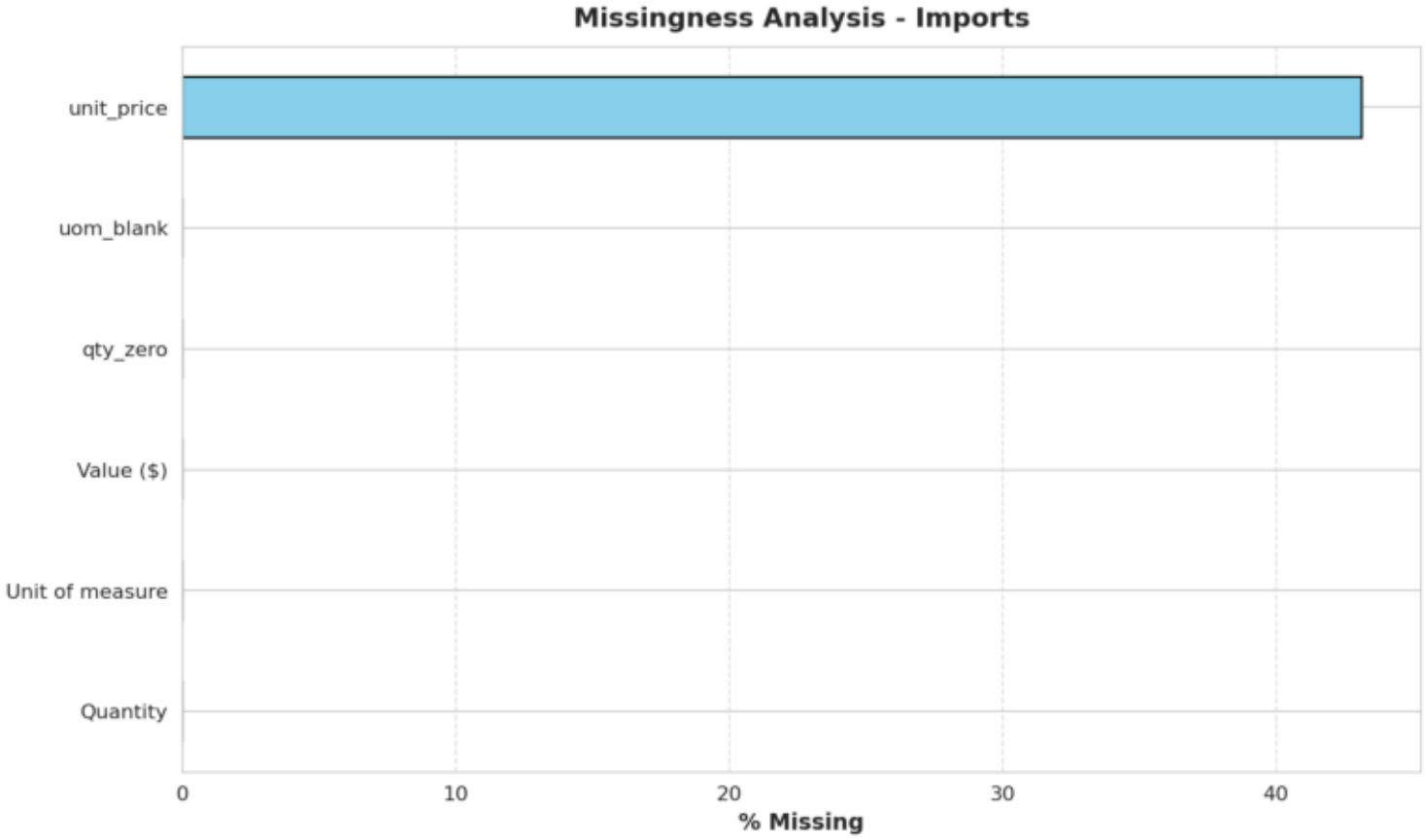
Unit of Measure Challenges

- 0 different commodities appear with multiple units
- This prevents accurate quantity summation and price comparison
- Kg, liters, units and tons are the most common conflicting units

What should we do?

→ Create and apply a standardization table: e.g. always convert to kg where possible

Export Data — Comparison & Specifics



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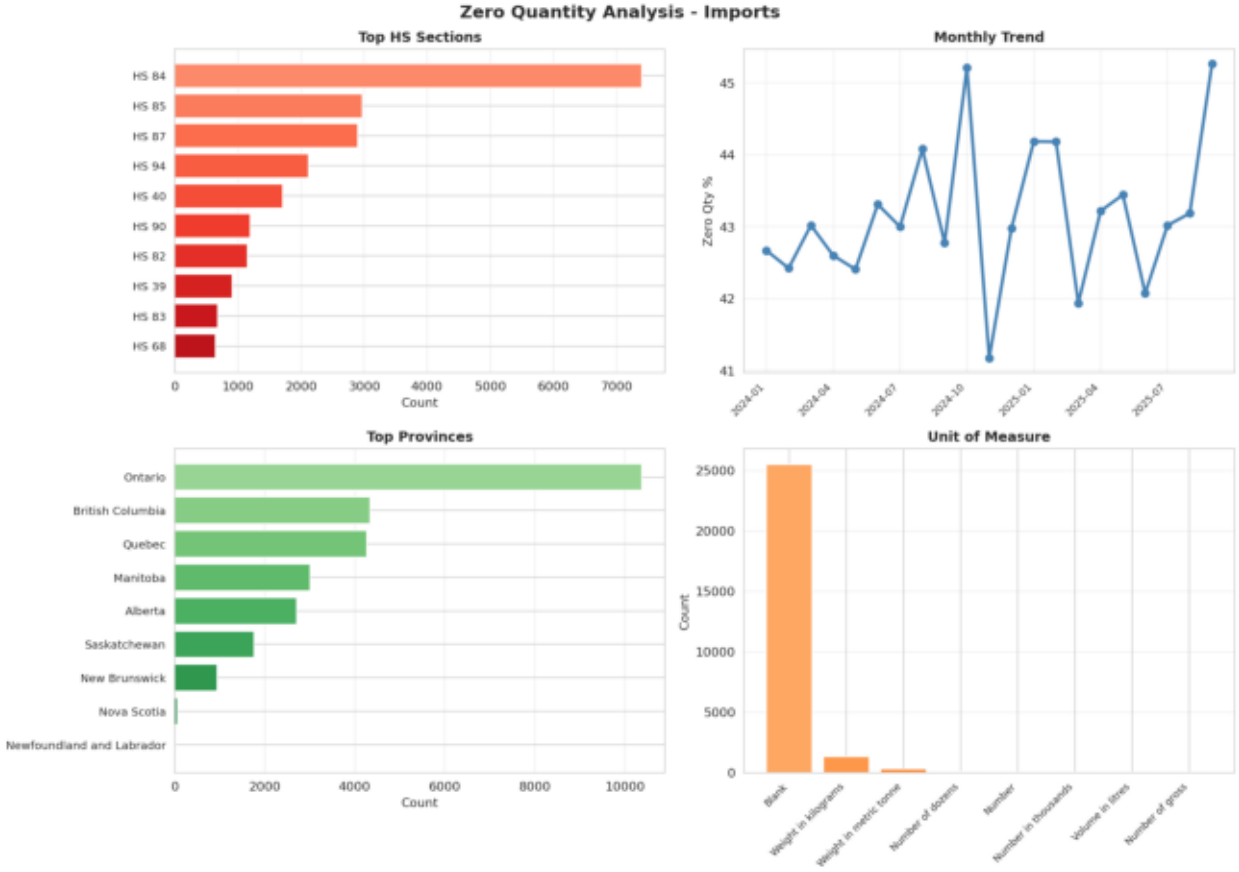

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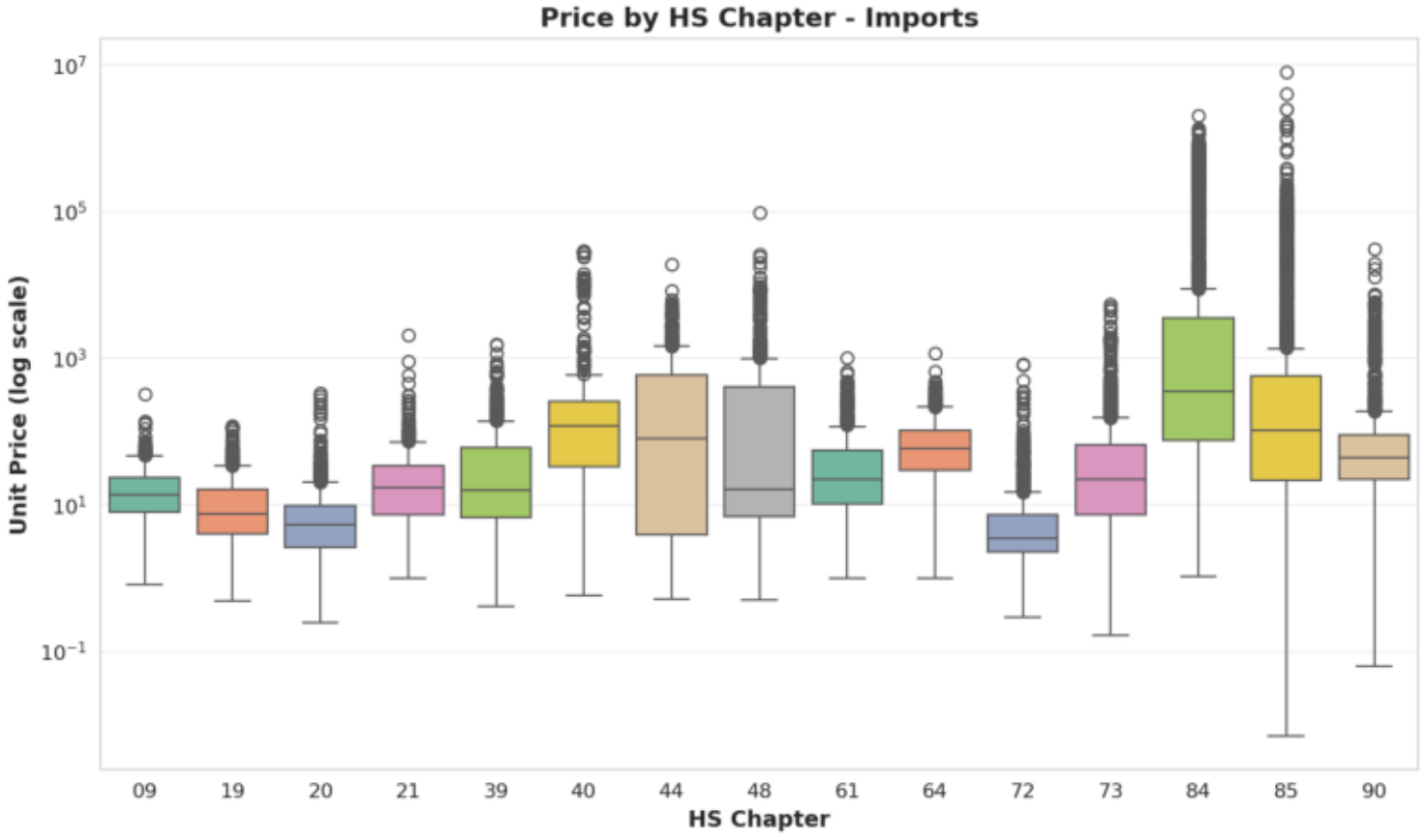

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Priority Actions & Next Steps

1. Zero-quantity records (highest impact)

→ Decide: impute / flag / exclude → implement in pipeline

2. Unit of measure standardization

→ Build HS-code → preferred unit mapping table

→ Apply conversion factors before any quantity aggregation

3. Price outlier treatment

→ Add is_outlier flag using HS-section IQR method

→ Review top outliers manually (especially > $10,000/unit)

4. Text normalization

→ Apply unidecode / NFKD normalization to Commodity, Province, Country

5. Final validation

Clean data = trustworthy trade insights

→ Re-run diagnostic after cleaning steps

→ Compare before/after aggregates (total value, top partners, etc.)